

Numpy (Matrix): Numpy is the library which handle multidimensional array (nd array)

n number of rows

n no of coloumns

Uses of Numpy :

1. Aerithmetic operations

2. Statistical operation

3. Bitwise operation

4. Copying & viewing arrays

5. Matrix operation

6. Linear algebra.

Numpy Crash course:

```
In [1]: import numpy as np
```

```
In [6]: np.__version__
```

```
Out[6]: '2.3.5'
```

```
In [7]: import sys  
sys.version
```

```
Out[7]: '3.13.9 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 19:09:58) [MSC v.1929 64 bit (AMD64)]'
```

Creating array

```
In [9]: my_list = [0,1,2,3,4,5]
```

```
In [10]: my_list
```

```
Out[10]: [0, 1, 2, 3, 4, 5]
```

```
In [11]: type(my_list)
```

```
Out[11]: list
```

coverting list to array

```
In [13]: np.array(my_list)
```

```
Out[13]: array([0, 1, 2, 3, 4, 5])
```

```
In [ ]: array([0,1,2,3,4,5])
```

```
In [19]: print(type(my_list))
```

```
<class 'list'>
```

```
In [22]: print(type(array))
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[22], line 1  
----> 1 print(type(array))  
  
NameError: name 'array' is not defined
```

```
In [25]: np.arange(10)
```

```
Out[25]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [26]: np.arange(10,20)
```

```
Out[26]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [28]: np.arange(10,50,5)
```

```
Out[28]: array([10, 15, 20, 25, 30, 35, 40, 45])
```

```
In [29]: np.arange(10,20,5,1)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[29], line 1  
----> 1 np.arange(10,20,5,1)  
  
TypeError: Cannot interpret '1' as a data type
```

```
In [30]: np.arange(-20,-10)
```

```
Out[30]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11])
```

```
In [31]: np.zeros(3)
```

```
Out[31]: array([0., 0., 0.])
```

```
In [32]: np.zeros(3, dtype = int)
```

```
Out[32]: array([0, 0, 0])
```

```
In [33]: np.zeros([2,2])
```

```
Out[33]: array([[0., 0.],
               [0., 0.]])
```

```
In [35]: np.zeros([10,10])
```

```
Out[35]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [36]: np.ones(4)
```

```
Out[36]: array([1., 1., 1., 1.])
```

```
In [37]: np.two(3)
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[37], line 1
----> 1 np.two(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'two'
```

```
In [38]: np.random.rand(3)
```

```
Out[38]: array([0.61299175, 0.29893667, 0.77491628])
```

```
In [39]: np.random.rand(3,4)
```

```
Out[39]: array([[0.02824779, 0.86152724, 0.32102138, 0.15796602],
               [0.24787851, 0.95961023, 0.61060489, 0.57743582],
               [0.20112156, 0.11398643, 0.64909358, 0.15751446]])
```

```
In [42]: np.random.randint(3)
```

```
Out[42]: 2
```

```
In [43]: np.random.randint(10,20)
```

```
Out[43]: 12
```

```
In [44]: np.random.randint(1,10,4)
```

```
Out[44]: array([6, 5, 6, 6], dtype=int32)
```

```
In [60]: m = np.random.randint(10,40,(10,10))    #
         print(m)
```

```
[[17 15 24 10 20 17 10 24 17 15]
 [11 35 22 21 10 10 34 36 35 28]
 [14 18 35 23 12 28 37 23 34 29]
 [23 34 10 17 13 29 26 32 26 14]
 [17 25 11 22 16 29 14 24 32 17]
 [34 11 12 38 30 28 27 21 33 39]
 [14 14 37 30 35 12 27 26 20 14]
 [18 17 24 30 10 26 21 24 13 10]
 [21 25 27 20 14 17 37 21 38 10]
 [36 21 19 30 34 32 36 35 35 25]]
```

```
In [68]: m = np.random.randint(1,10,(4,6))
         print(m)
```

```
[[9 3 1 4 1 6]
 [3 3 3 5 5 2]
 [4 2 7 6 8 6]
 [7 3 2 1 2 9]]
```

```
In [65]: m[0]
```

```
Out[65]: array([8, 2, 6, 4, 6, 6], dtype=int32)
```

```
In [74]: m = np.random.randint(1,10,(4,6))
         print(m)
```

```
[[7 3 6 4 7 9]
 [4 6 9 4 5 7]
 [7 8 9 4 2 1]
 [9 7 1 6 8 1]]
```

```
In [78]: print(m[-2,-3])
```

4

```
In [ ]:
```

```
In [ ]:
```